

Unit 2 Course Pack

Quantum States Algebra

Quantum Entanglement & Quantum Computing (UE23EC343AB3)
Photonics and Quantum Technology Lab, PES University

Machiraju Karthikeya, Rithvik Gurukishan
Teaching Assistants

Course Instructor: Dr. Kaustav Bhowmick

What Is In This Pack

Unit 2 of the syllabus, all fourteen hours, in eleven sections. **One section per syllabus topic, in syllabus order.**

Same structure as the Unit 1 pack:

1. a **blue box** giving the idea in plain words before any symbols;
2. the minimum mathematics needed to compute with;
3. **worked examples** in the same question-and-solution boxes as your question bank;
4. a **cyan problem box** with five problems.

Companion notebook: QEQC_Unit02_Lab.ipynb (Sections 10–11).

Where Unit 1 Left Off, and What Breaks

Unit 1 said: a state is a vector $|\psi\rangle$. That works when your system is isolated and you know exactly what was prepared.

Two very ordinary situations break it.

One. You do not know which state was prepared. A source flips a coin and sends $|0\rangle$ or $|1\rangle$, and nobody tells you the result. There is no vector for “ $|0\rangle$ or $|1\rangle$, I don’t know which.”

Two. You hold one half of an entangled pair. The pair as a whole is in a perfectly definite state, and yet *no vector at all* describes your half.

The fix for both is the same object: the **density matrix** ρ . Building it, and following where it leads, is the whole of Unit 2.

The Whole Unit In Five Lines

1. A **density matrix** ρ describes any state, known or not. Every prediction becomes $\text{tr}(\rho A)$.
2. Its **off-diagonal entries** carry the quantum-ness. Kill them and a superposition turns into an ordinary coin flip.
3. **Purity** $\text{tr}(\rho^2)$ tells you which case you are in. The Bloch sphere fills in to a solid ball.

4. **Tracing out** half of an entangled pair leaves you with a mixed state, even though the pair is perfectly known. That is EPR.
5. **Entropy** $S(\rho)$ measures how much you do not know, and for a pure pair it measures the entanglement.

Prerequisites

All of Unit 1, and in particular: Dirac notation, Hermitian and unitary operators, the trace, the Pauli matrices, the Bloch sphere, and the tensor product. Section 1 assumes you can compute $\langle \psi | A | \psi \rangle$ without pausing.

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1 Pure and Mixed States

The Idea

Here are two machines. Your job is to tell them apart.

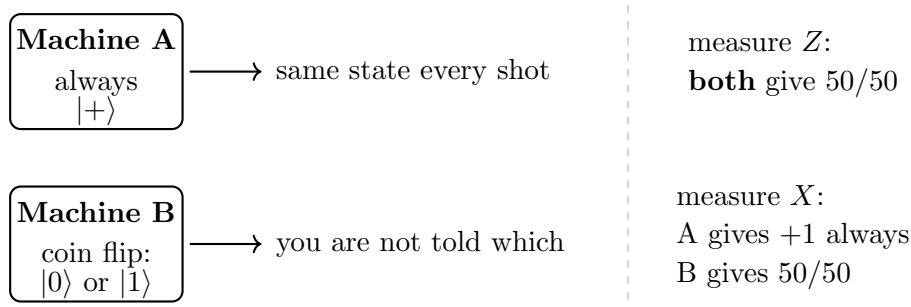
Machine A emits $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ every single time. Same state, every shot. Nothing random about it.

Machine B flips a fair coin. Heads it sends $|0\rangle$, tails it sends $|1\rangle$. It does not tell you the result.

Measure Z on either one and you get 50/50. Identical.

Now measure X . Machine A gives +1 **every time**. Machine B still gives 50/50.

So they are physically different machines, and yet **no single ket describes Machine B**. Writing $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ describes Machine A, and there is no vector meaning “ $|0\rangle$ or $|1\rangle$, I don’t know which.” We need a new object.



Pure and mixed

A **pure** state is one you can write as a single ket $|\psi\rangle$. Equivalently, as the projector

$$\rho = |\psi\rangle\langle\psi|.$$

A **mixed** state is a probabilistic list of pure states, $\{(p_i, |\psi_i\rangle)\}$, described by

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad p_i \geq 0, \quad \sum_i p_i = 1.$$

Two Different Kinds of Randomness

Both machines give random answers, but for different reasons.

Machine A is random because of the *geometry* of quantum mechanics. $|+\rangle$ is a state of complete knowledge, and it still gives a coin flip in Z because $|+\rangle$ sits at 90° to the Z axis on the Bloch sphere.

Machine B is random because of *your ignorance*. The qubit really is $|0\rangle$ or really is $|1\rangle$; you just were not told.

The entire job of the density matrix is to keep these two straight, and to tell you that only one number’s worth of difference between them is measurable.

Superposition Is Not a Mixture

This is the single most confused pair of objects in the course. Write both as matrices and the difference is visible:

$$\rho_A = |+\rangle \langle +| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \rho_B = \frac{1}{2} |0\rangle \langle 0| + \frac{1}{2} |1\rangle \langle 1| = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The **diagonals are identical**, which is why measuring Z cannot tell them apart. The **off-diagonals differ**, and those are exactly the relative phase from Unit 1. Those off-diagonal entries are called **coherences**. Interference lives in them. A mixture has already thrown them away.

1.1 Worked examples

Example 1.1. Write the density matrix for the state $|+\rangle$, and for an equal mixture of $|0\rangle$ and $|1\rangle$. Show that measuring Z cannot distinguish them but measuring X can.

Solution.

Pure state. With $|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$,

$$\rho_A = |+\rangle \langle +| = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Mixture.

$$\rho_B = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I}{2}.$$

Measuring Z . The probability of getting 0 is the top-left entry in both cases, $\frac{1}{2}$. Same for $|1\rangle$. Indistinguishable.

Measuring X . Use $P_+ = |+\rangle \langle +| = \rho_A$. For Machine A the probability of $+$ is 1 (the state already *is* $|+\rangle$). For Machine B,

$$p(+) = \langle + | \frac{I}{2} | + \rangle = \frac{1}{2}.$$

Why this matters. One measurement basis sees nothing; another separates them completely. That difference sits entirely in the off-diagonal entries.

Example 1.2. A source emits $|0\rangle$ with probability $\frac{3}{4}$ and $|1\rangle$ with probability $\frac{1}{4}$. Write its density matrix and compute $\langle Z \rangle$.

Solution.

$$\rho = \frac{3}{4} |0\rangle \langle 0| + \frac{1}{4} |1\rangle \langle 1| = \begin{pmatrix} 0.75 & 0 \\ 0 & 0.25 \end{pmatrix}.$$

For the expectation value, use $\langle Z \rangle = \text{tr}(\rho Z)$:

$$\rho Z = \begin{pmatrix} 0.75 & 0 \\ 0 & 0.25 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0.75 & 0 \\ 0 & -0.25 \end{pmatrix}, \quad \text{tr}(\rho Z) = 0.75 - 0.25 = 0.5.$$

Check. Directly, $\langle Z \rangle = (+1)(0.75) + (-1)(0.25) = 0.5$. ✓ The trace formula is doing ordinary weighted averaging.

Problems 1.1–1.5

1. Write the density matrix for $|-\rangle$ and confirm it differs from $|+\rangle\langle +|$ only in the sign of the off-diagonals.
2. Write ρ for an equal mixture of $|+\rangle$ and $|-\rangle$. What do you get, and why is that interesting?
3. Write ρ for the ensemble $\{\frac{1}{3}, |0\rangle\}, \{\frac{2}{3}, |+\rangle\}$.
4. For $\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|+\rangle\langle +|$, compute $\langle Z \rangle$ and $\langle X \rangle$.
5. Explain in two sentences why a mixture cannot produce interference.

2 Ensemble and Density Matrix

The Idea

Section 1 showed we need a new object. Here we build it, and it turns out to be forced on us rather than invented.

Suppose a source prepares $|\psi_i\rangle$ with probability p_i . What is the average of some measurement A over many runs? You have to average twice: once over the quantum randomness inside each state, and once over the classical randomness of which state was sent.

$$\langle A \rangle = \sum_i p_i \langle \psi_i | A | \psi_i \rangle.$$

Now do one line of algebra and watch a single matrix fall out of it, with the ensemble list nowhere in sight. That matrix is ρ .

The density matrix

Using $\langle \psi | A | \psi \rangle = \text{tr}(A | \psi \rangle \langle \psi |)$ from Unit 1,

$$\langle A \rangle = \sum_i p_i \text{tr}(A | \psi_i \rangle \langle \psi_i |) = \text{tr} \left[A \underbrace{\sum_i p_i | \psi_i \rangle \langle \psi_i |}_{\rho} \right].$$

So define

$$\rho = \sum_i p_i | \psi_i \rangle \langle \psi_i |, \quad \langle A \rangle = \text{tr}(\rho A).$$

Why This Is the Right Object

Look at where the ensemble list went. It appears nowhere in $\text{tr}(\rho A)$.

Everything you can ever predict is contained in ρ . The list of which states with which probabilities is extra information that no experiment can reach. That is not a defect of the formalism; Section 3 shows it is essential.

2.1 The three conditions

ρ is a valid state exactly when

- (i) $\rho^\dagger = \rho$ (Hermitian)
- (ii) $\rho \geq 0$, i.e. all eigenvalues ≥ 0 (positive semidefinite)
- (iii) $\text{tr} \rho = 1$ (unit trace)

Each Condition Earns Its Place

These are not arbitrary. Each one blocks a specific absurdity.

- **Hermitian**, so that $\langle A \rangle = \text{tr}(\rho A)$ comes out *real* for any observable A . Measurement results are real numbers.
- **Positive**, so that $\langle \phi | \rho | \phi \rangle \geq 0$ for every $|\phi\rangle$. That quantity is a probability, and there are no negative probabilities.
- **Unit trace**, so the probabilities add to 1.

2.2 Everything from Unit 1, rewritten for ρ

	Pure state $ \psi\rangle$	Any state ρ
the state	unit vector in $\mathcal{H}$	$\rho \geq 0, \text{tr} \rho = 1$
average of A	$\langle \psi A \psi \rangle$	$\text{tr}(\rho A)$
probability of outcome m	$ \langle a_m \psi \rangle ^2$	$\text{tr}(P_m \rho)$
state after that outcome	$\frac{P_m \psi\rangle}{\ P_m \psi\rangle\ }$	$\frac{P_m \rho P_m}{\text{tr}(P_m \rho)}$
evolution	$ \psi\rangle \rightarrow U \psi\rangle$	$\rho \rightarrow U \rho U^\dagger$

What to Remember

The right column *contains* the left as the special case $\rho = |\psi\rangle \langle \psi|$. Density matrices are not a competing theory. They are the general one, and kets are the lucky special case where you happen to know everything.

2.3 Coherences

For a qubit, write out the entries:

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}.$$

Populations and coherences

The diagonal entries ρ_{00}, ρ_{11} are the **populations**: the probabilities of measuring 0 and 1.

The off-diagonal entries $\rho_{01}, \rho_{10} = \rho_{01}^*$ are the **coherences**: they carry the relative phase, and they are what makes interference possible.

Decoherence is the process that shrinks the coherences toward zero while leaving the populations alone. A superposition slowly turns into an ordinary coin flip.

2.4 Worked examples

Example 2.1. A source prepares $|0\rangle$ with probability $\frac{1}{3}$ and $|+\rangle$ with probability $\frac{2}{3}$. Find ρ , verify the three conditions, and compute $\langle X \rangle$.

Solution.

$$\rho = \frac{1}{3} |0\rangle \langle 0| + \frac{2}{3} |+\rangle \langle +| = \frac{1}{3} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{2}{3} \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} + \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}.$$

Conditions. It is real and symmetric, so Hermitian. $\checkmark$ Trace = $\frac{2}{3} + \frac{1}{3} = 1$. $\checkmark$ For positivity, $\det \rho = \frac{2}{9} - \frac{1}{9} = \frac{1}{9} > 0$ and the trace is positive, so both eigenvalues are positive. $\checkmark$

Expectation.

$$\rho X = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}, \quad \langle X \rangle = \text{tr}(\rho X) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}.$$

Check. Averaging directly: $\frac{1}{3} \langle X \rangle_{|0\rangle} + \frac{2}{3} \langle X \rangle_{|+\rangle} = \frac{1}{3}(0) + \frac{2}{3}(1) = \frac{2}{3}$. $\checkmark$

Example 2.2. Show that the coherences of $\rho = |+\rangle \langle +|$ decay to zero under repeated dephasing, where dephasing acts as $\rho \rightarrow \frac{1}{2}\rho + \frac{1}{2}Z\rho Z$. What does ρ become?

Solution.

Start from $\rho = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and compute $Z\rho Z$:

$$Z\rho Z = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

Averaging the two:

$$\frac{1}{2}\rho + \frac{1}{2}Z\rho Z = \frac{1}{2} \left[\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \right] = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I}{2}.$$

The physical picture. The populations ($\frac{1}{2}$ and $\frac{1}{2}$) never moved. The coherences went straight to zero. The state started as Machine A from Section 1 and ended as Machine B, and no measurement can now recover the difference. That is decoherence in one calculation.

Problems 2.1–2.5

1. Find ρ for $\{\frac{1}{2}, |0\rangle\}$, $\{\frac{1}{2}, |+\mathbf{i}\rangle\}$ and check the three conditions.
2. Show that $\text{tr}(\rho A)$ is real for any Hermitian A and any valid ρ .
3. Show that $\rho \rightarrow U\rho U^\dagger$ preserves all three conditions.
4. For a general qubit ρ , show that $\rho_{00} + \rho_{11} = 1$ and that $\rho_{10} = \rho_{01}^*$.
5. A partial dephasing acts as $\rho \rightarrow (1-p)\rho + pZ\rho Z$. Show that the coherences are multiplied by $(1-2p)$ and the populations are untouched.

3 The Space of Density Matrices

The Idea

In Unit 1 you drew every pure qubit state as a point on the *surface* of a sphere. You may have wondered what is inside.

Now you know: the inside is the mixed states. A pure state sits on the skin. Mix in some ignorance and the point moves inward. At the very centre sits $I/2$, which is total ignorance.

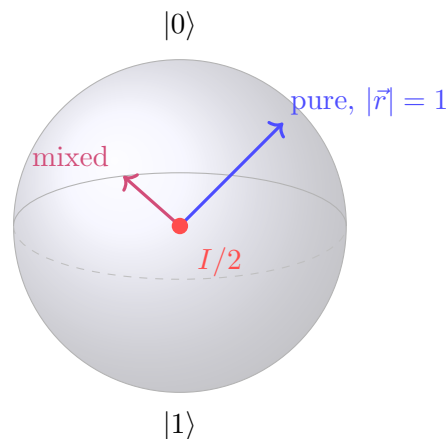
The sphere becomes a solid **ball**, and one number, the length of the arrow, tells you how much you know.

Bloch form of any qubit state

Any 2×2 Hermitian matrix with unit trace can be written

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma}) = \frac{1}{2} \begin{pmatrix} 1 + r_z & r_x - \mathbf{i}r_y \\ r_x + \mathbf{i}r_y & 1 - r_z \end{pmatrix}, \quad \vec{r} = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle).$$

Positivity forces $|\vec{r}| \leq 1$, so every state lives in the unit ball.



Purity: One Number Tells You Which Case You Are In

$$\mathcal{P}(\rho) = \text{tr}(\rho^2) = \frac{1}{2}(1 + |\vec{r}|^2).$$

- $\text{tr}(\rho^2) = 1 \iff \rho^2 = \rho \iff$ **pure**, on the surface.
- $\frac{1}{d} \leq \text{tr}(\rho^2) < 1 \iff$ **mixed**, inside.
- $\text{tr}(\rho^2) = \frac{1}{d} \iff$ **maximally mixed**, $\rho = I/d$, dead centre. For a qubit that is $\frac{1}{2}$. Purity is basis-independent, because the trace is. That makes it a genuine property of the state, not of how you chose to write it.

3.1 Convexity, and where pure states sit

Convexity

The set of density matrices is **convex**: if ρ_1 and ρ_2 are valid states, so is

$$\lambda\rho_1 + (1 - \lambda)\rho_2, \quad \lambda \in [0, 1].$$

On the Bloch ball this is literally the straight line between the two points: $\vec{r} = \lambda\vec{r}_1 + (1 - \lambda)\vec{r}_2$.

The **pure states are the extreme points**: the ones that cannot be written as a mixture of two different states. Geometrically, interior points lie on a chord between two others; surface points do not.

Two Consequences You Will Use Constantly

One. Mixing only ever moves you inward. No amount of classical randomness manufactures a pure state. You cannot make certainty out of coin flips.

Two. A gate rotates the ball; it never shrinks it. Unitary evolution $\rho \rightarrow U\rho U^\dagger$ preserves eigenvalues, hence preserves $\text{tr}(\rho^2)$ and $|\vec{r}|$.

Together these say: **no sequence of gates can undo decoherence**. Shrinking the ball takes a non-unitary process, and un-shrinking it needs measurement and fresh ancilla qubits. That is precisely why quantum error correction is built the way it is.

The Classical Picture Is a Segment, Not a Ball

A classical bit's states form a line segment with two corners. Every interior point has a **unique** decomposition into those two corners: if you know the mixture, you know the preparation.

The Bloch ball has a *continuum* of extreme points, and decompositions are wildly non-unique. That difference is the subject of the next box, and it turns out to be load-bearing.

Different Ensembles, Same ρ

$$\frac{1}{2} |0\rangle \langle 0| + \frac{1}{2} |1\rangle \langle 1| = \frac{I}{2} = \frac{1}{2} |+\rangle \langle +| + \frac{1}{2} |-\rangle \langle -|.$$

A source flipping a coin between $|0\rangle$ and $|1\rangle$, and a source flipping a coin between $|+\rangle$ and $|-\rangle$, produce **the same** ρ and are therefore **physically indistinguishable**. No measurement, ever, tells you which coin was flipped.

Is that a bug? No, and Section 5 shows why: if the ensemble were recoverable you could signal faster than light using half of an entangled pair. This ambiguity is exactly what enforces no-signalling.

3.2 Worked examples

Example 3.1. For $\rho = \frac{3}{4} |0\rangle \langle 0| + \frac{1}{4} |1\rangle \langle 1|$, find the Bloch vector, the purity, and state whether it is pure or mixed.

Solution.

$$\rho = \begin{pmatrix} 0.75 & 0 \\ 0 & 0.25 \end{pmatrix}.$$

Bloch vector. Compare with $\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$. The off-diagonals are zero, so $r_x = r_y = 0$. From the diagonal, $\frac{1}{2}(1 + r_z) = 0.75$, giving $r_z = 0.5$. So $\vec{r} = (0, 0, 0.5)$.

Purity.

$$\text{tr}(\rho^2) = 0.75^2 + 0.25^2 = 0.5625 + 0.0625 = 0.625.$$

Cross-check with the formula: $\frac{1}{2}(1 + 0.5^2) = \frac{1}{2}(1.25) = 0.625$. ✓

Verdict. $0.5 < 0.625 < 1$, so it is **mixed**, but not maximally. It sits partway up the z axis, leaning toward $|0\rangle$, which matches $p(0) = 0.75$.

Example 3.2. Show that no unitary can turn $\rho = I/2$ into a pure state.

Solution.

Under a unitary, $\rho \rightarrow U\rho U^\dagger$. Then

$$\text{tr}[(U\rho U^\dagger)^2] = \text{tr}[U\rho U^\dagger U\rho U^\dagger] = \text{tr}[U\rho^2 U^\dagger] = \text{tr}(\rho^2),$$

using $U^\dagger U = I$ and then cyclicity of the trace.

So purity is unchanged. For $\rho = I/2$ we have $\text{tr}(\rho^2) = \frac{1}{2}$, and a pure state needs $\text{tr}(\rho^2) = 1$. No unitary can bridge that gap.

Why this matters. Gates rotate the Bloch ball rigidly. If your qubit has decohered to the centre, no amount of clever circuitry recovers it. You need measurement, or a fresh qubit from the fridge.

Problems 3.1–3.5

1. Find $\vec{r}$ and $\text{tr}(\rho^2)$ for $\rho = |+\rangle\langle+|$. Confirm it is pure and locate it on the ball.
2. Show that $\text{tr}(\rho^2) \leq 1$ for every density matrix, with equality only for pure states. (Use the eigenvalues and $\sum_k \lambda_k = 1$.)
3. Verify $\text{tr}(\rho^2) = \frac{1}{2}(1 + |\vec{r}|^2)$ for a general qubit, and deduce $|\vec{r}| \leq 1$ from positivity.
4. Find two *different* three-state ensembles that both give $\rho = I/2$.
5. The depolarising channel acts as $\rho \rightarrow (1-p)\rho + p I/2$. Show that $\vec{r} \rightarrow (1-p)\vec{r}$, and find the purity after the channel for an initially pure state.

4 Cbits and Qbits

The Idea

Now that we have the full picture of a qubit's state space, we can line it up against a classical bit and see exactly what was gained.

The honest answer is subtler than the popular one. A qubit does hold a continuum of possible states, which sounds like infinite storage. But when you read it out you get **one bit**, and the state is destroyed.

So the advantage is not storage. It is that the amplitudes can *interfere* on the way to that one bit. Getting this straight now will save you from most of the nonsense written about quantum computing.

Side by side

	Cbit	Qubit
state space	segment $[0, 1]$	Bloch ball
definite states	2 (the endpoints)	a continuum (the surface)
reversible operations	2 (identity, NOT)	all of $SU(2)$, continuous
n units, parameters	$2^n - 1$ reals	$4^n - 1$ reals
readout	non-destructive	destructive, one bit per qubit
copying	free	forbidden (no-cloning)

The Information Paradox of the Qubit

A pure qubit state $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$ is set by two real numbers, which is infinitely many bits of specification.

And yet: **a single qubit can deliver at most one bit of classical information.** Measure it and you get one answer; the state is gone.

What to remember. The continuum of states is real but inaccessible. Quantum advantage never comes from “storing many numbers in the amplitudes.” It comes from arranging for those amplitudes to interfere so that a *classical* answer pops out at the end.

4.1 Registers

An n -qubit register lives in $(\mathbb{C}^2)^{\otimes n}$, dimension 2^n , with basis states $|x\rangle$ for every bit string $x \in \{0, 1\}^n$:

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} c_x |x\rangle, \quad \sum_x |c_x|^2 = 1.$$

The Superposition Trap

Applying H to every qubit of $|00 \cdots 0\rangle$ gives

$$H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_x |x\rangle,$$

all 2^n strings at once, from only n gates. It is tempting to call this “massive parallelism”.

But measure it and you get **one uniformly random string**, which you could have obtained from n coin flips. The superposition is worthless on its own. It is only useful if you then interfere the amplitudes so that the wrong answers cancel and the right one survives. That is what every algorithm in Unit 4 actually does.

4.2 Worked examples

Example 4.1. How many real parameters describe the state of n classical bits, and of n qubits (allowing mixed states)? Evaluate both for $n = 3$.

Solution.

Classical. A probability distribution over 2^n configurations needs 2^n numbers, minus one for normalisation:

$$2^n - 1.$$

For $n = 3$: $8 - 1 = 7$.

Quantum. A density matrix on $\mathcal{H}$ of dimension $d = 2^n$ is Hermitian (d^2 real parameters) with unit trace (minus one):

$$d^2 - 1 = 4^n - 1.$$

For $n = 3$: $64 - 1 = 63$.

Why this matters. Even at three qubits the quantum state space is nine times larger. The gap grows as 2^n , and it is the reason state tomography becomes impractical very quickly.

Example 4.2. A friend claims that because a qubit holds two real numbers, you could encode an entire textbook in a single qubit. Explain what is wrong.

Solution.

The encoding is fine. The **retrieval** is not.

You may certainly prepare $|\psi\rangle$ with θ and φ set to arbitrary precision. But to read the message you must measure, and a measurement on a single qubit returns **one of two outcomes** and destroys the state. One measurement, one bit.

You cannot repeat the measurement, because the state has collapsed. You cannot make copies first, because of no-cloning (Unit 1, Section 7). So one qubit yields at most one bit, however much you wrote into it.

In one sentence. The information is there; the laws of measurement will not let you out with more than one bit of it.

Problems 4.1–4.5

1. Compare $2^n - 1$ and $4^n - 1$ for $n = 1, 2, 5, 10$. Tabulate.
2. Write out $H^{\otimes 2} |00\rangle$ explicitly and confirm all four strings appear with equal amplitude.
3. Explain why measuring $H^{\otimes n} |0\rangle^{\otimes n}$ gives no computational advantage over classical coin flips.
4. How many complex amplitudes describe a pure state of 20 qubits? How much memory at 16 bytes each?
5. A classical bit can be copied freely. Give two consequences of the fact that a qubit

cannot.

5 The EPR Paradox

The Idea

Everything mixed so far came from a coin you did not look at: ordinary classical ignorance.

Now watch something much stranger. Take a two-qubit state that is **perfectly known**, completely pure, with no coin anywhere in the setup:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Ask what the first qubit alone is doing. The answer will be $I/2$: **maximally mixed**, total ignorance.

The whole is perfectly known. The parts are completely unknown. Classically that is impossible, and it is what Einstein, Podolsky and Rosen were objecting to.

5.1 The partial trace

Partial trace

To get the state of subsystem A alone, **trace out** B :

$$\rho_A = \text{tr}_B(\rho_{AB}) = \sum_j (I_A \otimes \langle j|_B) \rho_{AB} (I_A \otimes |j\rangle_B).$$

On product operators it is simply $\text{tr}_B(A \otimes B) = A \text{tr}(B)$.

This is the unique operation that reproduces every local measurement statistic on A . It is what “ignore B ” means mathematically.

The Calculation That Defines Unit 2

For $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$,

$$\rho_{AB} = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|).$$

Trace out B . The cross terms die because $\langle 0|1\rangle = 0$:

$$\text{tr}_B(|00\rangle\langle 11|) = |0\rangle\langle 1| \langle 1|0\rangle = 0.$$

What survives:

$$\rho_A = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I}{2}.$$

Read that again. We specified the pair completely. There was no ignorance anywhere. And qubit A on its own is in the state of *maximum possible ignorance*.

The signature of entanglement

A globally pure state whose reduced state is mixed is entangled. The more mixed the part, the more entangled the whole.

This is the second reason we needed density matrices, and it is the one that matters most. There is no vector describing qubit A ; only a ρ will do.

5.2 What EPR actually argued

The 1935 Argument, in Four Steps

EPR used two premises that sound completely reasonable.

Reality. If you can predict a quantity with certainty without disturbing the system, that quantity was already real.

Locality. Nothing Alice does can instantly affect anything at Bob's distant location. Now take the singlet $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, which gives *opposite* results in every basis.

1. Alice measures Z and gets $+1$. She now knows Bob will get -1 , with certainty, without touching his qubit. So Bob's Z value was real.
2. She could instead have measured X . By the same argument, Bob's X value was real.
3. But $[X, Z] \neq 0$, so quantum mechanics assigns no simultaneous values to both.
4. **Conclusion (EPR):** quantum mechanics is incomplete. There must be hidden variables.

How It Was Resolved

The EPR deduction is *valid*. Its premises are locality plus reality, taken together.

Bell later showed that those two premises, *as a pair*, force an inequality that quantum mechanics violates. Experiments then went and violated it.

So the conclusion fails because a premise fails, and the premise that fails is **local realism**. Unit 3 does the Bell inequality properly and you will test it on real hardware.

5.3 No signalling

Entanglement Does Not Send Messages

The correlations are instant, and this still does not let you signal.

No-signalling theorem. Nothing Alice does to her half changes Bob's reduced state ρ_B .

Why. If Alice applies any measurement $\{M_m\}$ and we do not condition on her result, Bob's state is

$$\sum_m \text{tr}_A[(M_m \otimes I)\rho(M_m^\dagger \otimes I)] = \text{tr}_A\left[\left(\sum_m M_m^\dagger M_m \otimes I\right)\rho\right] = \text{tr}_A[\rho] = \rho_B,$$

using completeness. Bob sees $I/2$ no matter what Alice does.

The correlations only become visible when Alice and Bob **compare notes over a classical channel**, and that travels at or below c . This is exactly why teleportation in Unit 3 needs two classical bits sent alongside.

Why the Ensemble Ambiguity Was Load-Bearing

Recall from Section 3 that $\frac{1}{2}\{|0\rangle, |1\rangle\}$ and $\frac{1}{2}\{|+\rangle, |-\rangle\}$ give the same $\rho = I/2$. Here is why that had to be true. If Alice measures Z on her half of $|\Phi^+\rangle$, Bob's qubit is left in the ensemble $\frac{1}{2}\{|0\rangle, |1\rangle\}$. If she measures X instead, Bob's is left in $\frac{1}{2}\{|+\rangle, |-\rangle\}$.

If Bob could tell those ensembles apart, he would learn Alice's choice of measurement basis instantly, at any distance. **Faster-than-light signalling.** The ambiguity you met in Section 3 as a curiosity is precisely what prevents it.

5.4 Worked examples

Example 5.1. Compute $\rho_A = \text{tr}_B(|\Psi^-\rangle\langle\Psi^-|)$ for the singlet $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, and show it is unchanged if Alice first applies an arbitrary unitary U_A .

Solution.

The reduced state.

$$\rho_{AB} = \frac{1}{2}(|01\rangle\langle 01| - |01\rangle\langle 10| - |10\rangle\langle 01| + |10\rangle\langle 10|).$$

Tracing out B : the cross terms give $\text{tr}_B(|01\rangle\langle 10|) = |0\rangle\langle 1| \langle 0|1\rangle = 0$, and the diagonal terms give

$$\rho_A = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I}{2}.$$

After Alice's unitary. Alice acting means $\rho_{AB} \rightarrow (U_A \otimes I)\rho_{AB}(U_A^\dagger \otimes I)$, so

$$\rho_A \rightarrow U_A \left(\frac{I}{2} \right) U_A^\dagger = \frac{1}{2} U_A U_A^\dagger = \frac{I}{2}.$$

Unchanged.

Why this matters. The maximally mixed state is the one state that every unitary leaves alone. Alice can do whatever she likes and her own marginal never budes, which is half the reason no-signalling holds.

Example 5.2. For $|\psi\rangle = \frac{1}{\sqrt{3}}|00\rangle + \sqrt{\frac{2}{3}}|11\rangle$, find ρ_A and its purity. Compare with $|\Phi^+\rangle$.

Solution.

Write $|\psi\rangle = a|00\rangle + b|11\rangle$ with $a = 1/\sqrt{3}$, $b = \sqrt{2/3}$. Then

$$\rho_{AB} = |a|^2 |00\rangle\langle 00| + ab^* |00\rangle\langle 11| + a^*b |11\rangle\langle 00| + |b|^2 |11\rangle\langle 11|.$$

The cross terms vanish under tr_B , leaving

$$\rho_A = |a|^2 |0\rangle\langle 0| + |b|^2 |1\rangle\langle 1| = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{2}{3} \end{pmatrix}.$$

Purity:

$$\text{tr}(\rho_A^2) = \frac{1}{9} + \frac{4}{9} = \frac{5}{9} \approx 0.556.$$

Comparison. For $|\Phi^+\rangle$, $\rho_A = I/2$ and the purity is 0.5, the minimum. Here it is 0.556, slightly higher.

What to remember. This state is entangled but *less* entangled than $|\Phi^+\rangle$. The more mixed the reduced state, the more entangled the pair. Section 6 turns that observation into a number.

Problems 5.1–5.5

1. Compute ρ_A for the product state $|+\rangle \otimes |0\rangle$. Is it pure or mixed? What does that tell you?
2. Compute ρ_A and ρ_B for $|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$.
3. Show that tr_B of any product state $\rho_A \otimes \rho_B$ returns ρ_A .
4. For $|\psi\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle$, find ρ_A and its purity as a function of θ . For which θ is it most entangled?
5. Explain in three sentences why the instant correlation in $|\Phi^+\rangle$ does not permit faster-than-light communication.

6 Density Matrices and Entropies

The Idea

Purity told you *whether* a state is mixed. Entropy tells you *how much*, in bits, and it is the version that connects to information theory.

The definition is Shannon's familiar $-\sum p \log p$, applied to the eigenvalues of ρ .

The payoff is at the end of the section: for an entangled pair, the entropy of one half **is** the amount of entanglement. A quantity invented to measure ignorance turns out to measure a physical resource.

6.1 Shannon entropy, in one box

Classical entropy

For a probability distribution $\{p_i\}$,

$$H(\{p_i\}) = - \sum_i p_i \log_2 p_i.$$

It is the average number of bits needed to say which outcome occurred. $H = 0$ for a certain outcome; $H = \log_2 n$ for n equally likely outcomes.

6.2 Von Neumann entropy

Quantum entropy

$$S(\rho) = - \text{tr} (\rho \log_2 \rho) = - \sum_k \lambda_k \log_2 \lambda_k,$$

where λ_k are the eigenvalues of ρ , with the convention $0 \log 0 = 0$.

In words: **diagonalise ρ , then apply Shannon's formula to the eigenvalues.**

Properties Worth Knowing

- (i) $S(\rho) \geq 0$, and $S = 0$ **exactly when** ρ is pure.
- (ii) $S(\rho) \leq \log_2 d$, with equality only for $\rho = I/d$. For a qubit, the maximum is 1 bit.
- (iii) Unchanged by unitaries: $S(U\rho U^\dagger) = S(\rho)$. Gates do not create or destroy ignorance.
- (iv) Additive on products: $S(\rho_A \otimes \rho_B) = S(\rho_A) + S(\rho_B)$.
- (v) Mixing increases it (concavity).

Read $S(\rho)$ as: **how many bits of classical ignorance you have about the system.**

Here Is Where Quantum Breaks Classical Intuition

Classically, the whole is always at least as uncertain as a part: $H(X, Y) \geq H(X)$. Knowing less about a part than about the whole is nonsense.

Quantum mechanically this **fails**. For $|\Phi^+\rangle$,

$$S(\rho_{AB}) = 0 \quad \text{but} \quad S(\rho_A) = 1 \text{ bit.}$$

The joint state is perfectly known while each half is maximally unknown.

There is no classical analogue of this, and it is arguably the sharpest single statement of what entanglement is.

6.3 Entanglement entropy

Measuring entanglement

For a *pure* bipartite state $|\psi\rangle_{AB}$, define

$$E(|\psi\rangle) = S(\rho_A) = S(\rho_B).$$

The two halves always give the same answer.

- $E = 0 \iff$ product state, no entanglement.
- $E = 1$ bit $\iff$ maximally entangled for two qubits, e.g. any Bell state.
- E cannot be increased by local operations and classical communication, which is what makes it a genuine *resource*.

One unit, $E = 1$ bit, is called an **ebit**.

6.4 Worked examples

Example 6.1. Compute $S(\rho)$ for (a) $\rho = |+\rangle\langle+|$, (b) $\rho = I/2$, and (c) $\rho = \text{diag}(0.75, 0.25)$.

Solution.

(a) **Pure.** Eigenvalues are 1 and 0:

$$S = -1 \log_2 1 - 0 \log_2 0 = 0 \text{ bits.}$$

(b) **Maximally mixed.** Eigenvalues both $\frac{1}{2}$:

$$S = -2 \left(\frac{1}{2} \log_2 \frac{1}{2} \right) = -2 \left(\frac{1}{2} \times (-1) \right) = 1 \text{ bit.}$$

(c) **Partially mixed.** Already diagonal, so the eigenvalues are 0.75 and 0.25:

$$S = -0.75 \log_2 0.75 - 0.25 \log_2 0.25 = -0.75(-0.415) - 0.25(-2) = 0.311 + 0.5 = 0.811 \text{ bits.}$$

What to remember. The scale runs from 0 (you know everything) to 1 (you know nothing), and 0.811 says this state is quite mixed but still leans toward $|0\rangle$.

Example 6.2. For $|\psi\rangle = \cos \theta |00\rangle + \sin \theta |11\rangle$, find the entanglement entropy as a function of θ , and evaluate it at $\theta = 0$ and $\theta = \pi/4$.

Solution.

From Section 5, tracing out B gives

$$\rho_A = \begin{pmatrix} \cos^2 \theta & 0 \\ 0 & \sin^2 \theta \end{pmatrix}.$$

It is already diagonal, so its eigenvalues are $\cos^2 \theta$ and $\sin^2 \theta$:

$$E(\theta) = -\cos^2 \theta \log_2 \cos^2 \theta - \sin^2 \theta \log_2 \sin^2 \theta.$$

At $\theta = 0$. $|\psi\rangle = |00\rangle$, a product state. Eigenvalues are 1 and 0, so $E = 0$. No entanglement, as expected.

At $\theta = \pi/4$. $|\psi\rangle = |\Phi^+\rangle$. Eigenvalues are both $\frac{1}{2}$, so $E = 1$ bit. Maximal.

The physical picture. E rises smoothly from 0 to 1 as θ goes from 0 to $\pi/4$. Entanglement is not on-or-off; there is a continuum, and this formula measures where on it you sit.

Problems 6.1–6.5

1. Compute $S(\rho)$ for $\rho = \frac{1}{2}(I + rZ)$ as a function of $r \in [0, 1]$, and sketch it. Confirm $S(0) = 1$ and $S(1) = 0$.
2. Show that $S(\rho) = 0$ implies ρ is pure.
3. Verify $S(U\rho U^\dagger) = S(\rho)$ for any unitary U .
4. Find E for $|\psi\rangle = \frac{1}{\sqrt{3}}|00\rangle + \sqrt{\frac{2}{3}}|11\rangle$.
5. Compute $S(\rho_{AB})$, $S(\rho_A)$ and $S(\rho_B)$ for $|\Psi^-\rangle$ and confirm that $S(\rho_{AB}) < S(\rho_A)$, which is classically impossible.

7 Superposition of States

The Idea

We have circled this several times. Now we settle it with the machinery in hand. Two questions get confused constantly:

- *Is this state a superposition?* — has no answer until you say **with respect to which basis**.
- *Is this a superposition or a mixture?* — has a definite, basis-independent answer, and ρ gives it to you.

The first is a statement about your description. The second is a statement about physical reality.

7.1 Superposition is basis-dependent

An Incomplete Question

$|+\rangle$ is a superposition in the Z basis. In the X basis it is a plain basis state, no superposition at all. The state did not change; the description did.

So asking “is this state in superposition?” is like asking “is this vector diagonal?”.

Always name the basis.

7.2 Superposition versus mixture is not basis-dependent

	Superposition $ +\rangle$	Mixture $\{ 0\rangle, 1\rangle\}$
what it is	one definite state	a coin flip over two states
randomness from	geometry of $\mathcal{H}$	your ignorance
before measurement	the system <i>is</i> $ +\rangle$	it <i>is</i> $ 0\rangle$ or $ 1\rangle$
ρ	$\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$	$\frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
purity	1	$\frac{1}{2}$
measure Z	50/50	50/50
measure X	+1 certain	50/50
interference	yes	no

Why This Matters

The last row is the one that counts. A superposition has **relative phase**, stored in the off-diagonal entries, and phase is what produces interference. A mixture has already thrown the phase away.

Notice that purity settles the question with a single number and never asks you to choose a basis. That is what makes it trustworthy.

If you have been saying “the system is in $|0\rangle$ or $|1\rangle$ but we don’t know which” about $|+\rangle$: stop. That sentence correctly describes a mixture and is wrong about a superposition.

7.3 Decoherence: how one turns into the other

Decoherence

Coupling a system to an environment that records information about it drives

$$\begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \longrightarrow \begin{pmatrix} \rho_{00} & 0 \\ 0 & \rho_{11} \end{pmatrix}.$$

Populations survive. Coherences die. A superposition becomes a classical probability distribution over the basis states.

The timescale for this is T_2 , and it is the number that decides how long a quantum computation can run.

The Which-Path Experiment, Explained at Last

In Unit 1 you ran a circuit where a single CNOT into an unread ancilla killed the interference fringes, and the explanation was “a record exists.” Now you can say it precisely.

The CNOT entangled the system with the ancilla. Tracing out the ancilla, which is what “we never looked at it” means mathematically, sends the off-diagonal entries of ρ to zero. No fringes.

Nothing mysterious happened. **Decoherence is just the partial trace of Section**

5 applied to an environment you cannot track.

7.4 Worked examples

Example 7.1. Two students each hand you a qubit. One prepared $|+\rangle$; the other flipped a coin between $|0\rangle$ and $|1\rangle$. You may make one type of measurement on many copies. What do you measure, and what do you expect to see?

Solution.

Measure X . That is the basis in which the two differ.

For the superposition, $\rho = |+\rangle\langle+|$, so

$$p(+) = \langle+|\rho|+\rangle = 1.$$

Every single shot gives $+$.

For the mixture, $\rho = I/2$, so

$$p(+) = \langle+|\frac{I}{2}|+\rangle = \frac{1}{2}.$$

Half the shots give $+$, half give $-$.

What not to do. Measuring Z gives 50/50 for both and tells you nothing, because the diagonals are identical.

Why this matters. There is no “universal” measurement. Choosing the basis *is* the experiment, and the right choice here is the one that reads the coherences.

Example 7.2. A qubit starts in $|+\rangle$ and dephases so that its coherences are multiplied by $\lambda \in [0, 1]$. Write $\rho(\lambda)$, find its purity and Bloch vector, and say what happens at $\lambda = 1$ and $\lambda = 0$.

Solution.

Populations unchanged at $\frac{1}{2}$ each, coherences scaled:

$$\rho(\lambda) = \frac{1}{2} \begin{pmatrix} 1 & \lambda \\ \lambda & 1 \end{pmatrix}.$$

Bloch vector. Comparing with $\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$, the off-diagonal is $\frac{1}{2}(r_x - ir_y)$, so $r_x = \lambda$, $r_y = 0$, and the equal diagonals give $r_z = 0$. So $\vec{r} = (\lambda, 0, 0)$.

Purity.

$$\text{tr}(\rho^2) = \frac{1}{2}(1 + |\vec{r}|^2) = \frac{1}{2}(1 + \lambda^2).$$

The two ends. At $\lambda = 1$: $\vec{r} = (1, 0, 0)$, purity 1, the state is $|+\rangle$ sitting on the surface. At $\lambda = 0$: $\vec{r} = \vec{0}$, purity $\frac{1}{2}$, the state is $I/2$ at the centre.

The physical picture. Dephasing slides the Bloch vector straight in along the x axis toward the centre. The state does not rotate anywhere; it just gets shorter, and shortness is exactly loss of quantum-ness.

Problems 7.1–7.5

1. Write ρ for $|+i\rangle$ and identify its coherences. In which basis does it look like a plain basis state?
2. Show that $|-\rangle\langle-|$ and the equal mixture of $|0\rangle, |1\rangle$ have the same Z statistics but different X statistics.
3. A state has $\rho = \frac{1}{2} \begin{pmatrix} 1 & 0.4 \\ 0.4 & 1 \end{pmatrix}$. Find its purity and entropy. Is it closer to a superposition or a mixture?
4. Explain, using the partial trace, why entangling a qubit with an environment destroys its coherences.
5. Is $|0\rangle$ a superposition? Answer properly.

8 Distinguishable and Indistinguishable States

The Idea

Someone hands you a qubit and says it is either $|\psi_0\rangle$ or $|\psi_1\rangle$. Which is it? If the two states are **orthogonal**, you can always tell, with certainty, every time. Just measure in a basis containing both. If they are **not orthogonal**, you can never be sure. Not with a cleverer detector, not with a better lab. It is forbidden by the structure of the theory. That limitation sounds like bad news, and it is the foundation of quantum cryptography.

8.1 Orthogonal states: always distinguishable

The easy case

If $\langle\psi_0|\psi_1\rangle = 0$, measure in a basis that contains both states. Outcome ψ_0 can only occur if the state was $|\psi_0\rangle$, and likewise for ψ_1 . Success probability: 1. No errors, ever.

8.2 Non-orthogonal states: never certain

The Helstrom bound

If $\langle \psi_0 | \psi_1 \rangle \neq 0$ and the two are equally likely, the best possible success probability over all measurements is

$$p_{\text{succ}} = \frac{1}{2} \left(1 + \sqrt{1 - |\langle \psi_0 | \psi_1 \rangle|^2} \right).$$

Better than guessing ($\frac{1}{2}$), worse than certain (1), and no measurement does better.

Why This Matters: BB84 in Three Sentences

Alice sends qubits chosen at random from $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$. These are not all mutually orthogonal: $|\langle 0 | + \rangle| = 1/\sqrt{2}$.

An eavesdropper cannot reliably determine which state was sent, and cannot copy it first (no-cloning), so any attempt to look introduces detectable errors.

Alice and Bob detect the eavesdropper by comparing a sample of their results. The security rests directly on the bound above.

What Is Actually Forbidden

Non-orthogonal states are not *indistinguishable*; they are *imperfectly* distinguishable. You get useful information, just never certainty.

And nothing here forbids distinguishing states you already know, or states drawn from a known orthogonal set. The restriction bites only for **unknown non-orthogonal** states, exactly as with no-cloning.

8.3 Consequences

Three Things That Follow

One. No cloning, again. If you could copy an unknown state you could make a thousand copies, measure each, and identify non-orthogonal states with certainty. Since that is impossible, cloning must be too. Unit 1 proved it from linearity; this is the information-theoretic version of the same fact.

Two. Secure key distribution. As above.

Three. A hard ceiling on readout. A qubit yields at most one bit, which is the Section 4 statement, now seen as a distinguishability limit.

8.4 A different meaning: identical particles

Two Uses of the Word “Indistinguishable”

(1) **Non-orthogonal states** that no measurement can reliably separate. That is everything above.

(2) **Identical particles** whose labels carry no physical meaning. For two identical particles the state must be symmetric under exchange (**bosons**, e.g. photons) or antisymmetric (**fermions**, e.g. electrons):

$$|\psi\rangle_{\text{boson}} \propto |a\rangle|b\rangle + |b\rangle|a\rangle, \quad |\psi\rangle_{\text{fermion}} \propto |a\rangle|b\rangle - |b\rangle|a\rangle.$$

Setting $a = b$ in the fermionic case gives zero: that is the Pauli exclusion principle. Bosonic symmetry gives photon bunching, which is the mechanism behind the entangled-photon sources you meet in Unit 3.

These two meanings are unrelated, and exam questions exploit the overlap. Read carefully.

8.5 Worked examples

Example 8.1. You are given one qubit, equally likely to be $|0\rangle$ or $|+\rangle$. Find the best possible probability of guessing correctly.

Solution.

First the overlap:

$$\langle 0|+\rangle = \frac{1}{\sqrt{2}}, \quad |\langle 0|+\rangle|^2 = \frac{1}{2}.$$

Now the Helstrom bound:

$$p_{\text{succ}} = \frac{1}{2} \left(1 + \sqrt{1 - \frac{1}{2}} \right) = \frac{1}{2} \left(1 + \frac{1}{\sqrt{2}} \right) = \frac{1}{2}(1.707) \approx 0.854.$$

What to remember. About 85%, not 100%. You do much better than guessing, and you can never be certain. On the Bloch sphere the two states are 90° apart rather than antipodal, and that gap is exactly what costs you the remaining 15%.

Example 8.2. Show that the Helstrom bound gives the right answers in the two extreme cases: orthogonal states, and identical states.

Solution.

Orthogonal. $\langle \psi_0 | \psi_1 \rangle = 0$, so

$$p_{\text{succ}} = \frac{1}{2} (1 + \sqrt{1 - 0}) = \frac{1}{2}(2) = 1.$$

Certainty, as it should be.

Identical. $|\langle \psi_0 | \psi_1 \rangle| = 1$, so

$$p_{\text{succ}} = \frac{1}{2} (1 + \sqrt{1 - 1}) = \frac{1}{2}.$$

Pure guessing, as it should be: there is nothing to tell apart.

In one sentence. The formula interpolates smoothly between the two cases you could have worked out without it, which is a good sign it is the right formula.

Problems 8.1–8.5

1. Find p_{succ} for distinguishing $|0\rangle$ from $\cos \theta |0\rangle + \sin \theta |1\rangle$, as a function of θ .
2. At what angle between two states does p_{succ} reach 0.95?
3. Explain why an eavesdropper who measures every BB84 qubit in the Z basis introduces errors, and estimate the error rate.
4. Two states have $|\langle \psi_0 | \psi_1 \rangle| = 0.3$. Find p_{succ} .
5. State clearly, in your own words, the difference between the two meanings of “indistinguishable” in this section.

9 Quantum Logic Gates

The Idea

A classical AND gate destroys information: output 0 tells you almost nothing about the input. That destruction has a physical cost, and it is why processors get hot. Quantum gates are unitary, so $U^{-1} = U^\dagger$ always exists. **Every quantum gate is reversible**, and in principle dissipates no heat at all.

This section collects the gates you will use for the rest of the course. Most of them appeared in Unit 1; what is new is the two-qubit gates and the question of which small set is enough to build everything.

Reversibility and Landauer’s Principle

Erasing one bit of information dissipates at least $k_B T \ln 2$ of heat. That is Landauer’s principle, and it is a statement about physics, not engineering.

Reversible gates erase nothing, so they escape that bound. Any classical computation can be made reversible using the Toffoli gate plus some ancilla bits, at the cost of carrying junk along.

Quantum computing inherits reversibility for free, because unitarity demands it.

9.1 Single-qubit gates

The standard set

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ (bit flip),} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \text{ (phase flip),} \quad Y = iXZ \text{ (both),}$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

With $S^2 = Z$, $T^2 = S$, $H^2 = I$, and $HXH = Z$.

9.2 Two-qubit gates

CNOT and friends

$$\text{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad |a, b\rangle \mapsto |a, a \oplus b\rangle.$$

The same pattern gives any controlled gate: $\text{C-}U = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$.

Two more you will meet:

$$CZ = (I \otimes H) \text{CNOT} (I \otimes H), \quad \text{SWAP} = \text{CNOT}_{12} \text{CNOT}_{21} \text{CNOT}_{12}.$$

CZ is symmetric between the two qubits; CNOT is not.

The Four Bell States

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \quad |\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle).$$

They form an orthonormal basis of $\mathbb{C}^4$. All four are maximally entangled, all four have $\rho_A = I/2$ and $E = 1$ ebit.

You get all four from $|\Phi^+\rangle$ by applying I , Z , X or XZ to just one qubit. That fact is the engine of dense coding and teleportation in Unit 3.

9.3 Universality

Which gates are enough?

$\{H, T, \text{CNOT}\}$ is **universal**: any unitary on any number of qubits can be approximated to arbitrary accuracy from these three.

$\{H, S, \text{CNOT}\}$, the **Clifford** group, is **not**. By the Gottesman–Knill theorem, circuits made only of Clifford gates can be simulated efficiently on a classical computer.

The T Gate Is Where the Advantage Lives

The difference between the two sets above is T versus S , a 45° rotation versus a 90° one. That small difference is the entire boundary between classically simulable and not.

Awkwardly, T is also the expensive gate to implement fault-tolerantly. A great deal of quantum architecture research is really about minimising the number of T gates in a circuit.

9.4 Worked examples

Example 9.1. Compute $\text{CNOT}(|+\rangle \otimes |-\rangle)$ and explain the surprise.

Solution.

Expand both qubits:

$$|+\rangle |-\rangle = \frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle) = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle).$$

CNOT flips the second bit wherever the first is 1, so $|10\rangle \rightarrow |11\rangle$ and $|11\rangle \rightarrow |10\rangle$:

$$\rightarrow \frac{1}{2}(|00\rangle - |01\rangle + |11\rangle - |10\rangle) = \frac{1}{2}(|0\rangle - |1\rangle)(|0\rangle - |1\rangle) = |-\rangle |-\rangle.$$

The surprise. The *control* qubit changed, from $|+\rangle$ to $|-\rangle$, even though CNOT nominally acts on the target. The target is unchanged.

This is called **phase kickback**, and it is the mechanism behind Deutsch–Jozsa, Bernstein–Vazirani, phase estimation and Shor. Understand it now and Unit 4 becomes much easier.

Example 9.2. Verify that applying Z to the first qubit of $|\Phi^+\rangle$ produces $|\Phi^-\rangle$, and that the reduced state ρ_A is unchanged.

Solution.

The state.

$$(Z \otimes I) \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = |\Phi^-\rangle.$$

The Z leaves $|0\rangle$ alone and flips the sign of $|1\rangle$, so only the second term picks up a minus.

The reduced state. Tracing out B from $|\Phi^-\rangle$, the cross terms vanish exactly as before and the signs sit only on those cross terms, so

$$\rho_A = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I}{2},$$

identical to $|\Phi^+\rangle$.

Why this matters. Alice changed the global state to an orthogonal one using a local gate, and Bob’s marginal did not move at all. That is no-signalling again, and it is

what makes dense coding work: the information is stored in the correlations, not in either qubit.

Problems 9.1–9.5

1. Show that the four Bell states are mutually orthogonal.
2. Verify $CZ = (I \otimes H) \text{CNOT} (I \otimes H)$ by matrix multiplication.
3. Apply X to the first qubit of $|\Phi^+\rangle$. Which Bell state do you get?
4. Show that $\text{SWAP} = \frac{1}{2}(I \otimes I + X \otimes X + Y \otimes Y + Z \otimes Z)$.
5. Explain why $\{H, S, \text{CNOT}\}$ cannot be universal, given the Gottesman–Knill theorem.

10 Introduction to the IBM and Microsoft Quantum Environments

The Idea

Everything in this unit is checkable in a few lines of code, and from here on you should be checking it.

Two ecosystems dominate. **IBM Quantum** works through Qiskit, a Python library, and runs on IBM’s superconducting hardware. **Microsoft Azure Quantum** uses Q#, a dedicated language, and dispatches to partner hardware.

This course uses Qiskit. The concepts transfer directly, and nothing in this section depends on the choice.

The two ecosystems

	IBM Quantum / Qiskit	Azure Quantum / Q#
language	Python library	Q#, a dedicated language
style	circuit-first	program-first, classical control
hardware	IBM superconducting transmons	partners (IonQ, Quantinuum, ...)
simulator	Aer	full-state and sparse simulators
entry point	quantum.ibm.com	Azure portal / VS Code

10.1 Setup

```
# pip install qiskit qiskit-aer numpy
import numpy as np
from qiskit import QuantumCircuit, transpile
from qiskit.quantum_info import Statevector, DensityMatrix, partial_trace,
    entropy
from qiskit_aer import AerSimulator
```

```
np.set_printoptions(precision=4, suppress=True)
```

10.2 Lab 1: building a state and reading its density matrix

```
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
print(qc.draw())

sv = Statevector.from_instruction(qc)
print(sv.data) # [0.707, 0, 0, 0.707] -> |Phi+>

rho = DensityMatrix(sv)
print(rho.data)
print("purity =", rho.purity()) # 1.0 -> the pair is pure
```

10.3 Lab 2: the partial trace, in five lines

This is Section 5 executed.

```
rho_A = partial_trace(rho, [1]) # trace out qubit 1
print(rho_A.data) # [[0.5, 0], [0, 0.5]] -> I/2
print("purity =", rho_A.purity()) # 0.5 -> maximally mixed
print("entropy =", entropy(rho_A, base=2)) # 1.0 -> one ebit
```

Read Those Three Numbers

Global purity 1. Local purity 0.5. Local entropy 1 bit.

That is the numerical fingerprint of a maximally entangled pure state, and it is the whole EPR argument in three lines of output. The pair is perfectly known; each half is maximally unknown.

10.4 Lab 3: superposition versus mixture, numerically

```
plus = np.array([1, 1], dtype=complex) / np.sqrt(2)
rho_pure = np.outer(plus, plus.conj()) # [[.5,.5],[.5,.5]]
rho_mix = np.eye(2, dtype=complex) / 2 # [[.5, 0],[ 0,.5]]

X = np.array([[0, 1], [1, 0]], dtype=complex)
Z = np.array([[1, 0], [0, -1]], dtype=complex)

for name, r in [("pure ", rho_pure), ("mixed", rho_mix)]:
    print(name,
          "<Z> =", np.real(np.trace(r @ Z)),
          "<X> =", np.real(np.trace(r @ X)),
          "purity =", np.real(np.trace(r @ r)))
# pure <Z> = 0.0 <X> = 1.0 purity = 1.0
# mixed <Z> = 0.0 <X> = 0.0 purity = 0.5
```

Identical in Z , completely different in X . That is Section 1's two machines, now as output.

Problems 10.1–10.5 (computational)

1. Build all four Bell states with two-gate circuits and verify their pairwise orthogonality using `Statevector.inner`.
2. For each Bell state, compute ρ_A and confirm it is $I/2$.
3. Prepare $|\psi\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle$ using `ry` and `cx`, then plot $S(\rho_A)$ against $\theta \in [0, \pi/2]$. Compare with the analytic curve from Section 6.
4. Write a function that takes any ρ and returns its Bloch vector, and test it on $|+\rangle$, $|+i\rangle$ and $I/2$.
5. Verify numerically that $\text{tr}(\rho^2) = \frac{1}{2}(1 + |\vec{r}|^2)$ for several random qubit states.

11 Simulation of Quantum Circuits on IBM Quantum

The Idea

A simulator is perfect. Real hardware is not.

Running the same circuit in both places is the fastest way to understand what decoherence actually costs you, and it turns Sections 3 and 7 from theory into something you have measured yourself.

The Bell circuit should give only 00 and 11. On real hardware you will also see 01 and 10, a few percent of the time. This section is about where those come from.

11.1 Running on the simulator

```
qc = QuantumCircuit(2, 2)
qc.h(0)
qc.cx(0, 1)
qc.measure([0, 1], [0, 1])

sim = AerSimulator()
counts = sim.run(transpile(qc, sim), shots=4096).result().get_counts()
print(counts) # about {'00': 2050, '11': 2046}; no '01' or '10'
```

11.2 Running on real hardware

```
# requires a free IBM Quantum account
from qiskit_ibm_runtime import QiskitRuntimeService, SamplerV2

service = QiskitRuntimeService()
backend = service.least_busy(operational=True, simulator=False)

isa = transpile(qc, backend=backend, optimization_level=3)
job = SamplerV2(mode=backend).run([isa], shots=4096)
print(job.result()[0].data.c.get_counts())
```

What Real Results Look Like

You will see nonzero counts for 01 and 10, typically a few percent. These come from readout error, two-qubit gate infidelity, and decoherence during the circuit.

Do not treat this as a bug in your code. Pull T_1 , T_2 and the two-qubit gate error from `backend.properties()` and check whether the error rate you see is consistent with your circuit depth. That comparison is the actual exercise.

11.3 Adding noise on purpose

Rather than waiting in a hardware queue, you can dial noise in yourself.

```
from qiskit_aer.noise import NoiseModel, depolarizing_error

nm = NoiseModel()
nm.add_all_qubit_quantum_error(depolarizing_error(0.02, 1), ['h'])
nm.add_all_qubit_quantum_error(depolarizing_error(0.05, 2), ['cx'])

noisy = AerSimulator(noise_model=nm)
print(noisy.run(transpile(qc, noisy), shots=4096).result().get_counts())
```

Connect It Back to Section 3

Sweep the depolarising error rate p upward and watch the Bell correlations degrade. Then compute the Bloch vector length of the reduced state at each p . It should follow $\vec{r} \rightarrow (1 - p)\vec{r}$, exactly the contraction of the Bloch ball you met in Section 3.

Why this matters. Noise is not a vague nuisance. It is a specific, measurable shrinking of the state space, and you can plot it.

11.4 Worked examples

Example 11.1. You run the Bell circuit on hardware and get $\{ '00': 1950, '11': 1900, '01': 130, '10': 116 \}$ out of 4096 shots. Estimate the error rate and compute the observed purity of the two-qubit state assuming the errors are depolarising.

Solution.

Error rate. The forbidden outcomes are 01 and 10:

$$\frac{130 + 116}{4096} = \frac{246}{4096} \approx 0.060.$$

About 6% of shots landed where the ideal circuit forbids.

Purity estimate. Model the state as $\rho = (1 - p) |\Phi^+\rangle \langle \Phi^+| + p \frac{I}{4}$. The depolarised part spreads over all four outcomes equally, so it contributes $p/2$ to the two forbidden ones:

$$\frac{p}{2} = 0.060 \implies p \approx 0.12.$$

Then

$$\text{tr}(\rho^2) = (1-p)^2 + \frac{2p(1-p)}{4} + \frac{p^2}{4} \approx 0.774 + 0.053 + 0.004 \approx 0.83.$$

What to remember. The ideal state has purity 1. At 6% leakage you are already down to roughly 0.83, i.e. noticeably inside the surface. Errors show up as loss of purity long before they make the output unusable.

Example 11.2. A circuit has depth 20 with gate time 200 ns, on a device with $T_2 = 80 \mu\text{s}$. Estimate the coherence loss over the circuit, and say whether the circuit is feasible.

Solution.

Circuit duration.

$$t = 20 \times 200 \text{ ns} = 4000 \text{ ns} = 4 \mu\text{s}.$$

Coherence remaining. Coherences decay as e^{-t/T_2} :

$$e^{-4/80} = e^{-0.05} \approx 0.951.$$

So roughly 95% of the coherence survives, i.e. about 5% is lost to dephasing alone.

Verdict. Feasible. The rule of thumb is $t \ll T_2$, and here $t/T_2 = 1/20$.

Think of it this way. A circuit of depth 400 would give $t = T_2$ and leave only $e^{-1} \approx 37\%$ of the coherence, which is where results stop meaning anything. Circuit depth budget is really a coherence-time budget, and gate errors add on top of this.

Problems 11.1–11.5 (computational)

1. Run the Bell circuit on the ideal simulator with 100, 1000 and 10000 shots. How does the deviation from 50/50 scale?
2. Sweep the depolarising error on `cx` from 0 to 0.2 and plot the fraction of forbidden outcomes.
3. For each error rate above, compute ρ_A and its purity. Confirm the Bloch contraction $\vec{r} \rightarrow (1-p)\vec{r}$.
4. Pull T_1 , T_2 and the `cx` error from a real backend using `backend.properties()` and estimate the maximum useful circuit depth.
5. Run the Bell circuit on real hardware and compute the observed purity of ρ_A . Compare with the ideal value of 0.5 and explain the direction of the discrepancy.

Unit 2 Summary

Formula Sheet

Density matrix. $\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|$; Hermitian, $\rho \geq 0$, $\text{tr } \rho = 1$; $\langle A \rangle = \text{tr}(\rho A)$; $\rho \rightarrow U\rho U^\dagger$.

Pure vs mixed. Pure $\iff \rho^2 = \rho \iff \text{tr}(\rho^2) = 1$. Maximally mixed: $\rho = I/d$, $\text{tr}(\rho^2) = 1/d$.

Bloch ball. $\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$, $|\vec{r}| \leq 1$; $\text{tr}(\rho^2) = \frac{1}{2}(1 + |\vec{r}|^2)$; surface = pure, centre = $I/2$.

Coherences. Off-diagonal entries of ρ . Decoherence kills them and leaves populations alone.

Partial trace. $\rho_A = \text{tr}_B(\rho_{AB})$; $\text{tr}_B(A \otimes B) = A \text{tr}(B)$. Pure global state with mixed reduced state $\Rightarrow$ entangled.

Entropy. $S(\rho) = -\text{tr } \rho \log_2 \rho = -\sum_k \lambda_k \log_2 \lambda_k$; $0 \leq S \leq \log_2 d$; $S = 0 \iff$ pure.

Entanglement entropy. $E = S(\rho_A) = S(\rho_B)$ for a pure pair; $E = 1$ ebit for a Bell state.

The quantum surprise. $S(\rho_{AB}) < S(\rho_A)$ is possible. Classically it is not.

Distinguishing states. Orthogonal: certain. Otherwise $p_{\text{succ}} = \frac{1}{2}(1 + \sqrt{1 - |\langle \psi_0 | \psi_1 \rangle|^2})$.

Bell states. $|\Phi^\pm\rangle, |\Psi^\pm\rangle$: orthonormal basis, all maximally entangled, all with $\rho_A = I/2$.

Gates. CNOT = $|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X$; $\{H, T, \text{CNOT}\}$ universal; Clifford alone is classically simulable.

No signalling. Local operations never change the other party's ρ .

Six Mistakes That Cost Marks

1. **Superposition vs mixture.** Check the off-diagonals, or just compute the purity.
2. **Claiming the ensemble is recoverable from ρ .** It is not, and that fact enforces no-signalling.
3. **Forgetting that unitaries preserve purity.** No gate sequence undoes decoherence.
4. **Saying entanglement transmits information.** It does not; you always need a classical channel too.
5. **Using $S(\rho_{AB}) \geq S(\rho_A)$.** That is the classical inequality and it fails here.
6. **Confusing the two senses of “indistinguishable”,** non-orthogonal states versus identical particles.

Section to Syllabus Map

Syllabus topic	Section
Pure and mixed states	1
Ensemble and density matrix	2
Space of density matrices	3
Cbits and Qbits	4
EPR paradox	5
Density matrices and entropies	6
Superposition of states	7
Distinguishable and indistinguishable states and consequences	8
Quantum logic gates	9
Introduction to IBM/Microsoft quantum computer environment	10
Simulation of quantum circuits on IBM Quantum environment	11

What Comes Next

Unit 2 ended on a cliffhanger it did not resolve. The state $|\Phi^+\rangle$ is perfectly known, yet each half is maximally mixed. EPR concluded from this that the theory must be incomplete.

Unit 3 settles it. Bell's inequality turns "is there a local hidden variable?" into a number you can measure, and experiment gives the answer. Along the way you get sources of entangled photons, the no-cloning theorem properly stated, dense coding, teleportation, and quantum error correction, all built on the partial trace and entropy you learned here.

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